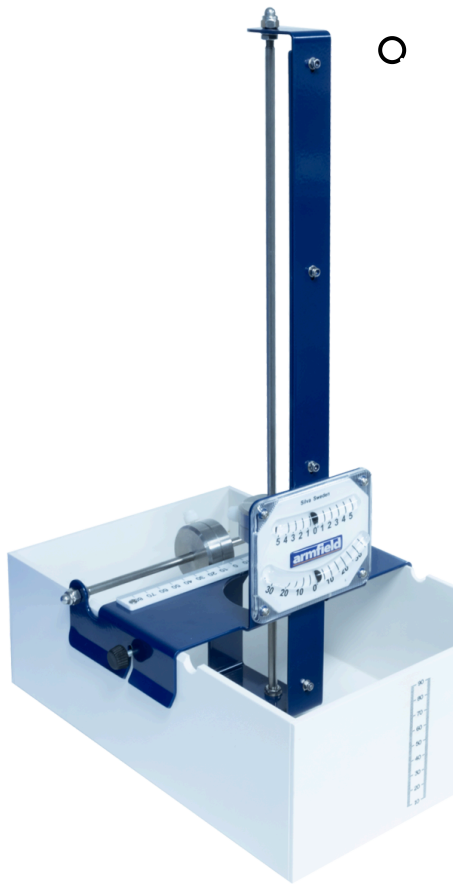




F1-14 MKII Metacentric Height

This equipment enables a thorough investigation of the factors affecting the stability of a floating body. The metacentre will produce stable and unstable equilibrium, a plumb-bob is suspended from a mast, indicating the angle of heel on a calibrated scale.

PRODUCT CODE: F1-14

Categories: Educational, F1-14 - Metacentric Height, IFT Industrial Food Technology

Tags: Angle Of Heel , Armfield , F1-14 , Fluid Mechanics , Hydraulics Bench , Metacentric Height , Stable , Unstable Equilibrium , Volumetric Tank

DESCRIPTION

The position of the metacentre can be varied to produce stable and unstable equilibrium.

The equipment consists of a plastic rectangular floating pontoon where the centre of gravity can be varied by an adjustable weight which slides and can be clamped in any position on a vertical mast.

A single plumb bob is suspended from the mast which indicates the angle of heel on a calibrated scale. A weight with lateral adjustment enables the degree of heel to be varied and hence the stability of the pontoon determined.

The equipment does not require a separate water tank as it may be used on the hydraulics bench by filling the volumetric tank.

TECHNICAL SPECIFICATIONS

Max angle of heel: $\pm 13^\circ$

Corresponding linear dimension: $\pm 90\text{mm}$

FEATURES & BENEFITS

Experimental Content

- Determining the centre of gravity of the pontoon
- Determining the metacentric height and from this the position of the metacentre for the pontoon

- Varying the metacentric height with angle of heel

Dimensions

Length: 0.35m

Width: 0.20m

Height: 0.475m

Order Codes

F1-14

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Armfield Limited. Unit 10 Headlands Business Park, Salisbury Road, Ringwood, Hampshire BH24 3PB.
England. Tel: +44(0) 1425 478781